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Original article

Association of fluoroquinolone resistance with rare quinolone resistance-determining region (QRDR) mutations and protein-quinolone binding affinity (PQBA) in multidrug-resistant *Escherichia coli* isolated from patients with urinary tract infection

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Background: Urinary tract infections (UTIs) caused by *Escherichia coli* pose significant public health risks, particularly in developing countries like Bangladesh. This study aimed to elucidate resistance patterns among UTI isolates and comprehensively investigate the mutational spectrum and its impact on drug-microbe interactions.
Methods: We collected and identified *E. coli* isolates from hospitalized UTI patients at Dhaka Medical College Hospital and determined their resistance patterns using the disc diffusion method and broth microdilution. Quinolone resistance-determining regions (QRDRs) of the target genes (*gyrA*, *gyrB*, *parC*, and *parE*) associated with fluoroquinolone resistance were amplified by polymerase chain reaction (PCR) and analyzed through BTSeq™ sequencing for mutations, followed by molecular docking analysis using PyMOL and AutoDock for the protein-quinolone binding affinity (PQBA) study.
Results: All isolates (100 %) displayed multidrug resistance, with chloramphenicol (16 % resistant) and colistin (28 % resistant) demonstrating superior efficacy compared to other antibiotics. The isolates resistant to colistin, as determined by disc diffusion testing, exhibited remarkably high minimum inhibitory concentrations (MICs), with one isolate registering an MIC exceeding 512 µg/mL. Alarming resistance rates were observed for five antibiotic classes, except for polymyxins (28 % resistant) and protein synthesis inhibitors (48 % resistant). Fifty-two percent (52 %) of the isolates exhibited resistance to all five tested quinolones. Sequence analysis revealed a novel L88Q mutation in ParC, affecting PQBA and binding conformation. Additionally, three ParC mutations (S80I, E84V, and E84G) and two ParE mutations (S458A and I529L) were identified, which had not been previously reported in Bangladesh. Among these, S80I appeared in all isolates. Double-mutations (S83L+D87N) in GyrA, L88Q and S80I in ParC, and I529L in ParE were identified as key drivers of fluoroquinolone resistance.
Conclusion: Our findings underscore the accumulation of significant mutations within QRDRs of UTI isolates, potentially compromising fluoroquinolone efficacy. The emergence of these novel mutations warrants further investigation to impede their dissemination and combat quinolone resistance.

Introduction

Multidrug-resistant Urinary tract infections (MDR UTIs) are the types of infections that do not respond to three or more classes of antimicrobials [1,2]. *Escherichia coli* is the predominant causative bacteria for UTIs and recovering from such infections largely depends on the efficacy of administered antimicrobials [3,4]. The fast

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evolution and spread of antimicrobial resistance mechanisms and limited treatment options have put UTIs in masses at a global risk. Previous research indicates that UTI isolates in Bangladesh have a substantially more severe resistance profile than isolates from other sources, such as sewage water or community settings [5,6]. According to a recent study at the International Centre for Diarrhoeal Disease Research in Bangladesh (icddr,b), 46 % of Gram-positive bacteria and 71 % of Gram-negative UTI isolates were MDR [3]. Eighty-five percent (85 %) of the bacterial isolates collected from 142 urine samples of Bangladeshi UTI patients were found to be colistin-resistant, which is thought to be the last resort for treating fetal infectious diseases, and all of those who were colistin-resistant also had the MDR phenomenon [7]. In a separate study on extended-spectrum beta-lactamase (ESBL)-producing bacteria obtained from young adult women attending government medical colleges in Bangladesh, a recent study concluded that 72 % of isolates were MDR and 39 % were resistant to third-generation antibiotics [8]. Due to the previously reported high incidence of MDR isolates in UTI patients in Bangladesh, we chose the busiest national hospital of Bangladesh, Dhaka Medical College Hospital, to study the β-lactam resistance pattern as well as to investigate the mutational spectrum responsible for quinolone resistance in UTI isolates.

Uropathogenic *Escherichia coli* (UPEC), the leading cause of UTIs, is growing increasingly resistant to antibiotics, including β-lactams, macrolides, colistin, and fluoroquinolones [7,9]. The development of serine β-lactamase enzymes is one of the ways that bacteria can develop resistance to β-lactams. Although serine β-lactamases and β-lactams have a covalent interaction, the enzymes can deacetylate this complex by using the antibiotic as a substrate. One method by which bacteria can gain resistance to β-lactams is the production of serine β-lactamase enzymes. Bacterial resistance to β-lactams can be caused by the enzymatic hydrolysis of the β-lactam bond within the drug molecule's β-lactam ring. The medication loses its antibacterial effectiveness when the β-lactam ring is disrupted. The most frequent method of β-lactam resistance, β-lactamases, mediate this route of resistance [10]. Additionally, bacteria can create a variety of antibiotic-defensive mechanisms. For example, gene alterations and variable expression of particular genes can enable bacteria to disregard, pump out, or escape the antibiotic [2,9].

Macrolides, for example, azithromycin and erythromycin, are among the least effective antibiotics against UPEC, which acts as a reservoir for spreading resistance genes [11]. Three mechanisms enable bacteria to resist macrolide antibiotics: (1) target-site alteration by methylation or mutation that inhibits the antibiotic from binding to its ribosomal target, (2) antibiotic efflux, and (3) drug inactivation [12]. Chromosome-encoded pumps help Gram-negative bacteria develop innate resistance to hydrophobic drugs like macrolides [12].

Fluoroquinolones are considered first-line drugs for treating UTIs [13,14]. However, the substantial prevalence of *E. coli* resistant to fluoroquinolones had recently been a significant concern in prescribing this class of antimicrobials [15]. Their widespread use has accompanied the emergence of resistance to fluoroquinolones. This resistance is primarily attributable to chromosomal mutations in the Quinolone resistance-determining regions (QRDRs) of the genes that encode the subunits of DNA gyrase and topoisomerase IV, for example, *gyrA*, *gyrB*, *parC,* and *parE*, protein products of which are the drugs' target enzymes [16]. The QRDRs refer to precise segments within the DNA gyrase and topoisomerase IV genes susceptible to genetic mutations, potentially altering the protein-antibiotic interaction and consequently reducing the susceptibility to fluoroquinolones [17].

Fluoroquinolones have been strategically engineered to discern the configuration of said genes and are proficient in instigating a potent interaction that undermines the inherent functionality of these genes and proteins [18]. The efficacy of fluoroquinolone antibiotics is contingent upon their capacity to engage with these enzymes and impede their operation [19]. Mutations in the genes that encode these enzymes can modify the protein-antibiotic interplay, which can subsequently reduce the vulnerability of bacteria to fluoroquinolones [18,20].

The significance of the Protein-quinolone binding affinity (PQBA) lies in its influence on the efficacy of quinolones in combating bacterial infections [21,22]. Antibiotics that exhibit a greater PQBA are capable of efficaciously suppressing bacterial growth by hindering the synthesis of crucial proteins within the bacterial organism, which are DNA gyrase and topoisomerases, in this case [22,23]. Fluoroquinolone antibiotics' effectiveness is impacted by their binding affinity towards proteins, which ultimately determines the strength of their bond with the target protein [18].

Our study evaluated the efficacy of five (5) different fluoroquinolones of the first, second, and fourth generations. We investigated the impact on PQBA of newly identified and previously documented mutations in QRDRs of *E. coli* isolated from Bangladeshi UTI patients. This study addresses critical knowledge gaps in fluoroquinolone resistance mechanisms in E. coli isolates from a resource-limited setting, identifying novel mutations and providing public health insights relevant to Bangladesh and similar regions.

Materials and methods

Sample collection

E. coli isolates were collected from the urine samples of the UTI patients who attended Dhaka Medical College Hospital (DMCH) between March 01, 2022, and July 31, 2022. The Ethical Review Committee of the Department of Biochemistry and Molecular Biology, University of Dhaka, approved the study (Ref. No. BMBDU-ERC/EC/22/005). In addition, we confirm that all methods used in this study followed the World Health Organization (WHO) "Annex 2, TRS 961" guidelines. Participants were informed about the nature of the research and experimental procedures. Informed written consent was obtained from all the study subjects before the samples were collected. This study exclusively enrolled hospital outpatients devoid of a hospitalization history. There were no age or gender restrictions in the sample selection. One hundred (100) patients were employed in the study following the strict inclusion and exclusion criteria, with 52 % male and 48 % female. Each participant was categorized into one of four age groups (≤18, 19–40, 41–60, and ≥61), where the highest proportion of sample donors fell within the 19–40 age group (Supplementary Table 1). Any bacterial isolates other than *E. coli* and morphologically similar isolates were excluded from the analysis. Additionally, patients currently undergoing antibiotic treatment were ineligible for participation.

Inclusion and exclusion criteria

Samples with fewer than 10^5 colony-forming units per milliliter *(cfu/mL)* growth were excluded from this study. Parameters such as white blood cell (WBC) count, positive urine culture, dysuria (painful or difficult urination), urgency to urinate, frequency of urination, lower abdominal pain or discomfort, hematuria (presence of blood in urine), and the characteristics of urine such as cloudiness or foul smell were considered to ascertain the presence of active infections and are essential inclusion criteria. Samples were collected from patients who met these criteria and exhibited colony counts surpassing the established threshold of 10^5 cfu/mL were chosen for further study.

Selective isolation and identification

E. coli was isolated and cultured according to the procedure described previously [24]. Specifically, urine samples were cultured on nutrient agar to isolate colonies, followed by subculturing on MacConkey agar and Eosin Methylene Blue (EMB) agar, where *E. coli* was identified by its characteristic pink and green-metallic sheen, respectively. *E. coli* isolates were also identified using a combination of biochemical tests (Analytical profile indexing (API) using API20E Biochemical Test Strips, BioMérieux, France), phenotypic methods (modified Hodge test, Gram staining, Chromogenic Coliform Agar (CCA) plating), and genotypic confirmation (16S rRNA sequencing) for representative isolates.

Antibiotic susceptibility analysis

Antibiotic susceptibility testing was conducted according to the protocol described in our previous studies [25,26]. Specifically, each UTI isolate underwent three iterations of susceptibility testing using the Kirby-Bauer disc diffusion method to measure the zone of inhibition (ZOI) diameters. Fifteen antibiotics from seven classes were tested (Supplementary Table 2), focusing on fluoroquinolone resistance with five specific fluoroquinolone drugs. Isolates were lawned on Mueller Hinton Agar (MHA) plates, inoculated with antibiotic discs, and incubated overnight at 37°C after culturing in Mueller Hinton Broth (MHB) for two hours, adjusting microbial growth to the 0.5 McFarland standard.

To assess colistin resistance, isolates showing presumed resistance via disc diffusion underwent broth microdilution assays to determine MIC. The interpretation of ZOI and MIC data adhered to the Clinical and Laboratory Standards Institute (CLSI-2022) and European Committee on Antimicrobial Susceptibility Testing (EUCAST-2022) guidelines. Out of 100 *E. coli* isolates, 25 exhibited multidrug resistance during initial testing.

Evaluation of carbapenemase activity

Meropenem-resistant isolates identified by disc diffusion were further evaluated for carbapenemase activity using the Modified Hodge test, with *Escherichia coli* ATCC 25922 as a control strain. A “cloverleaf” inhibition zone pattern indicated carbapenemase production.

Determination of MIC of fluoroquinolones

Fluoroquinolone resistance was determined via both disk diffusion and 12-point gradient broth microdilution to determine MIC, with results interpreted according to CLSI guidelines, providing reliable data for identifying resistant isolates selected for further antibiogram assessments and genetic analyses and to assess the contribution of mutations in resistance development.

DNA extraction and resistance gene screening

The investigation into the prevalence of five beta-lactamase genes and the QRDRs of four fluoroquinolone resistance genes within all isolates was conducted using PCR, followed by gel electrophoresis. The PCR-amplified DNA fragments were visualized after the extraction of bacterial genomic DNA utilizing a boiling DNA extraction protocol. Primer sequences necessary for amplifying the target segments of the QRDRs of DNA gyrase and topoisomerase genes (*gyrA, gyrB, parC,* and *parE*), along with narrow and extended-spectrum beta-lactamase genes (bla_{NDM}, $bla_{\text{CTX-M1}}$, bla_{TEM}, bla_{SHV1}, and bla_{VIM1}), were manufactured by Macrogen, Inc. (Supplementary Table 3). A negative control was included in each batch of PCR. The amplicons were resolved in 1.0 % agarose gel at 75 V for 60 minutes and visualized in the Alpha Imager HP Gel documentation system.

DNA sequencing and mutational analysis

Nine isolates met the stringent criteria for sequence analysis, namely, exhibiting resistance to all five fluoroquinolones tested and harboring all four fluoroquinolone resistance genes. These selected isolates underwent Sanger sequencing to elucidate the nucleotide-level mutations within the QRDRs of the DNA gyrase (*gyrA, gyrB*) and topoisomerase IV (*parC, parE*) genes. Sequencing was performed by Celemics, Inc., using the BTSeq™ Contiguous Sequencing program. The sequences have been made publicly accessible online on the National Center for Biotechnology Information (NCBI) and are associated with the following accession numbers: PP118286-PP118294 *(gyrA)*, PP133156-PP133164 *(gyrB)*, PP133165-PP133173 *(parC)*, and PP133174-PP133182 *(parE)*. Geneious Prime 2023 software was used to carry out a mutational analysis. All four genes were aligned with the *Escherichia coli* (INSCD-U00096) annotated reference gene sequence obtained from the NCBI Genome Browser, and an appropriate reading frame was chosen to prevent an internal stop codon. Any deviation from the reference sequence was referred to as a “point mutation,” and any point mutation that modified the amino acid sequence was referred to as a “missense mutation.”

Mutational impact analysis by molecular docking

The severity of the missense mutations’ effects on antibiotic-protein interactions was determined. The RCSB Protein Data Bank (PDB) (https://www.rcsb.org/) has been used to obtain the 3D structure of the wild-type protein, and SWISS-MODEL (https://swissmodel.expasy.org/) has been used to predict the 3D structure of the mutated proteins. Using AutoDock v4 [27], it was possible to predict the ideal sub-molecular binding site, the optimal protein-antibiotic interaction poses, and the affinity of both the wild-type and mutant proteins for all five antibiotics of the fluoroquinolone class. PyMol (version 2.5.4) [28] was used to visualize the protein-antibiotic interaction.

Statistical analysis

All statistical analyses were performed using GraphPad Prism (GraphPad Software, USA) and R (R Foundation for Statistical Computing, Austria). Using the one-way T-test, fluoroquinolones’ MICs were compared between isolates with and without specific QRDR mutations. A p-value of < 0.05 was considered statistically significant.

Results

Confirmation of the presence of Escherichia coli in patients with UTI

All the selected isolates exhibited a green metallic sheen on EMB agar, providing a preliminary indication of their identity as *E. coli*. Subsequent gram staining confirmed the isolates to be Gram-negative, rod-shaped bacteria, consistent with the morphology of *E. coli*. Notably, all isolates displayed the typical biochemical profile (as presented by Kligler’s Iron Agar, MR-VP, Citrate, Indole, oxidase, and catalase tests) associated with *E. coli*, confirming their identity. Further confirmation of isolates according to the automated API analyzer revealed the comprehensive biochemical profile of isolates, which is presented in Supplementary Table 4. Furthermore, the chromogenic properties of the selected isolates on CCA agar, as depicted in Supplementary Figure 1, were consistent with other observations, providing additional validation of their identity as *E. coli*. This collective evidence from multiple biochemical tests and

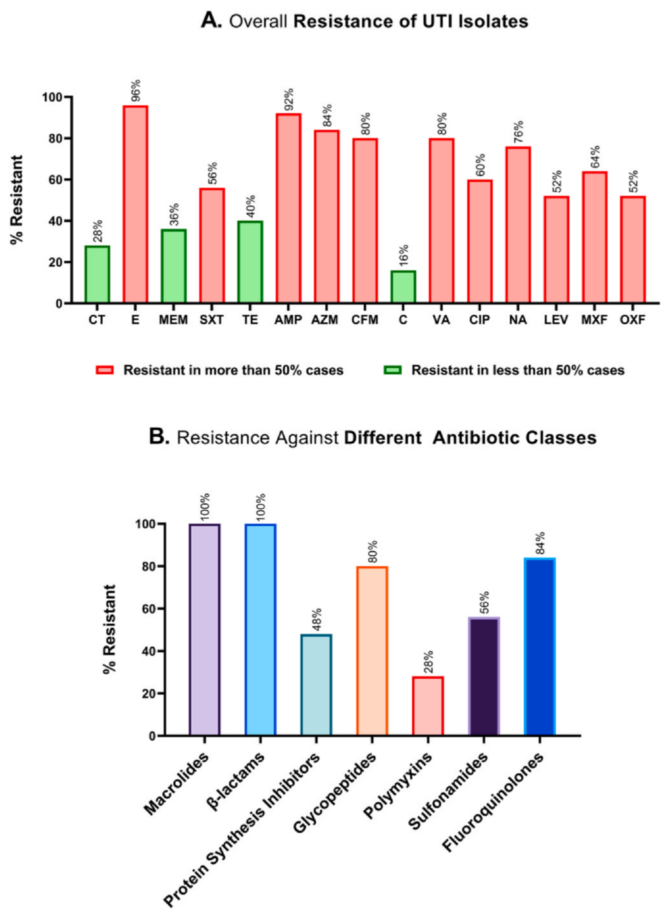

Fig. 1. Antibiotic resistance pattern of the clinical *E. coli* isolates collected from UTI patients. A. Resistance pattern against each of the 15 antibiotics. Red bars indicate the state of resistance in more than 50 % of the isolates, whereas green indicates less than 50 %. The remaining 11 antibiotics were less effective in killing or inhibiting microbial growth, except for chloramphenicol, colistin, meropenem, and tetracycline. CT: colistin; E: erythromycin; MEM: meropenem; SXT: Sulfamethoxazole-trimethoprim; TE: tetracycline; AMP: ampicillin, AZM: azithromycin; CFM: cefixime; C: chloramphenicol; VA: vancomycin; CIP: ciprofloxacin; NA: nalidixic acid; LEV: levofloxacin; MXF: moxifloxacin; OXF: ofloxacin. B. Resistance pattern against seven different antimicrobial classes. Besides polymyxins and protein synthesis inhibitors, other classes like macrolides, β-lactams, glycopeptides, sulfonamides, and fluoroquinolones have been alarmingly ineffective against clinical *E. coli* isolates.

chromogenic characteristics confirms accurately identifying the isolates as *E. coli*.

Resistance against different antibiotics

Antibiotic resistance in UTI patients was alarmingly high for most of the 15 antibiotics used in our study. First-generation antibiotics like erythromycin and ampicillin remain the least effective antimicrobials against UTI isolates, with a resistance percentage of 96 % and 92 %, respectively. Likewise, azithromycin (84 %), vancomycin (80 %), nalidixic acid (76 %), and sulfamethoxazole-trimethoprim (56 %) resistance have evolved catastrophically. Despite being a third-generation antibiotic, cefixime resistance (80 %) parallels some first-generation antibiotics, as per our findings. Resistance against all the quinolones appeared alarmingly high, including moxifloxacin (64 % resistant), a fourth-generation antibiotic. However, the efficacy of chloramphenicol (16 % resistant), colistin (28 % resistant), meropenem (36 % resistant), and tetracycline (40 % resistant) showed substantially better sensitivity towards each of the UTI isolates (Fig. 1A).

Resistance against different antibiotic classes

Antibiotics of different classes used in this study are summarized in Supplementary Table 2. For the classes having multiple representative antibiotics, resistance against at least one has been counted as resistant to the respective class. Macrolides and β-lactams were the least sensitive of those classes, with 100 % of the resistant isolates for both (Fig. 1B). These two classes of antibiotics are thus almost ineffective against any UTI *E. coli* isolates. Antimicrobial resistance against fluoroquinolones (84 % resistant isolates), glycopeptides (80 % resistant), and sulfonamides (56 % resistant) were alarmingly high, as observed in this study. Protein synthesis inhibitors were found to be slightly better with 48 % resistant isolates, whereas 12 % of the isolates have developed resistance against the antibiotics of this group, chloramphenicol and tetracycline. Polymyxin was most sensitive to all, but only 28 % of UTI isolates could render colistin of this class ineffective (Fig. 1B). Isolates exhibiting resistance to colistin in the disc diffusion method displayed notably high MICs, with one isolate recording a MIC surpassing 512 µg/mL. According to the CLSI guidelines (2022), a MIC of > 2 µg/mL indicates resistance of a specific isolate to colistin. Furthermore, all meropenem-resistant isolates demonstrated the ability to produce carbapenemase, as evidenced by positive results from the modified Hodge test (Supplementary Figure 2).

Multidrug resistance in clinical isolates

MDR refers to an isolate being resistant to three or more classes of antibiotics [29], as defined by CLSI M100 and EUCAST 2022. As illustrated in Fig. 2A, 100 % of the isolates were resistant to at least three classes of antibiotics, making the bacterial library completely MDR. An isolate resistant to all but two classes of antimicrobials is termed extensively drug-resistant (XDR); 60 % of our studied samples align with this criteria. Isolate U2 (Fig. 2B, red) has developed resistance against all the antibiotics tested, making it a potential pandrug-resistant (PDR) [30]. Thirty-six percent (36 %) of the UTI isolates were resistant to all three of the *β*-lactam antibiotics (Fig. 2C), and 100 % of the isolates were resistant to at least one *β*-lactam antibiotic, whereas 52 % of the isolates were resistant to all five fluoroquinolones and 16 % to none (Fig. 2D). The analysis of beta-lactamase gene prevalence revealed that bla_{NDM} was the most prevalent gene (76 %), followed by bla_{TEM} (64 %) and bla_{VIM1} (64 %). $bla_{\mathrm{CTX\text{-}M1}}$ showed a prevalence of 60 %, while bla_{SHV1} exhibited the lowest prevalence at 56 %. This distribution pattern (Fig. 2E) suggests a bla_{NDM} predominance among the studied population's beta-lactamase genes.

QRDR mutations in gyrA, parC, and parE genes

Quinolone resistance predominantly relies on mutations acquired in the QRDRs of the *gyrA, gyrB, parC,* and *pares* genes. Our study observed a high prevalence of the *gyrA*, *gyrB*, *parC*, and *parE* genes among the isolates. Notably, *gyrA* was detected in 84 % of isolates, whereas both *gyrB* and *parC* were present in 76 %, and *parE* was identified in 69.5 %. Mutations in the amplified QRDRs have been analyzed following BTSeq™ sequencing, and the results are summarized in Table 1. We have identified two missense mutations in GyrA, S83L, and D87N with 100 % and 88.9 % frequency, respectively. Four mutations have been identified in ParC, each with a very low frequency except for S80I, which has a 100 % frequency of occurrence. Lastly, two missense mutations have been noted in the *parE* gene with relatively low frequency. No missense mutation was identified in the amplified region of *gyrB*. Except for S83L and D87N, no other mutations, namely S80I, E84V, and E84G in ParC and S458A and I529L in ParE have been

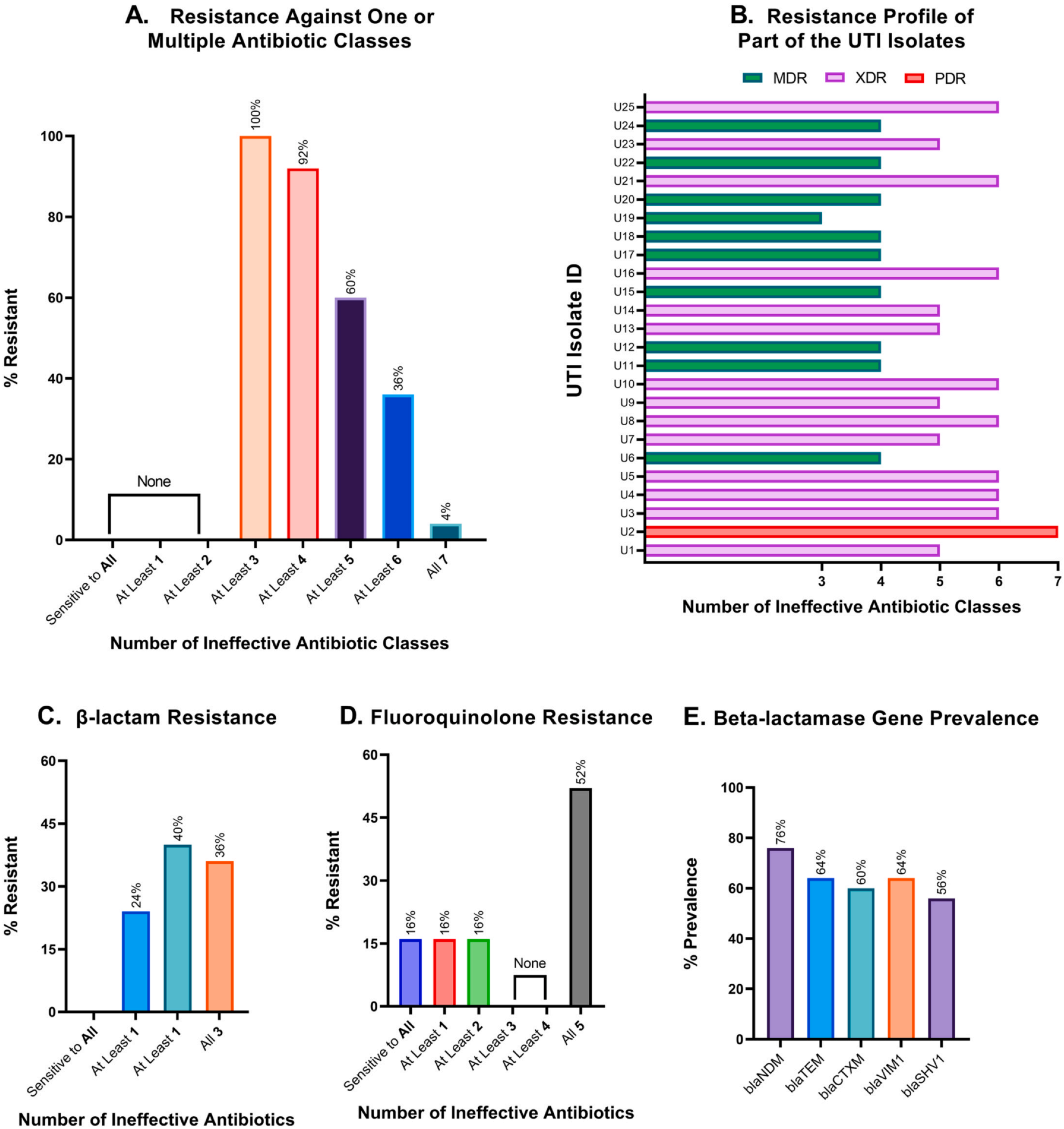

Fig. 2. Multidrug-resistance profiling of UTI isolates. *A.* Antimicrobial resistance against one or multiple classes of antibiotics. No isolate was resistant to less than three classes of antibiotics, and 36 % were resistant to at least six different classes of antibiotics. One isolate (4 %) was resistant to all 15 antibiotics of 7 classes. *B.* The level of resistance of each UTI isolates against the number of antibiotic classes. MDR: Multidrug-resistant; XDR: Extensively drug-resistant; PDR: Pandrug-resistant. *C.* Resistance against one or multiple β-lactams. No isolate was sensitive to all three β-lactam antibiotics (ampicillin, meropenem, cefixime). *D.* Resistance against one or multiple quinolones. Over half of the isolates have developed resistance against all five antibiotics of this class, whereas 16 % were sensitive to all five fluoroquinolones. *E.* Percent prevalence of five beta-lactamase genes in UTI isolates. All genes causing resistance to β-lactam antibiotics are found in at least 56 % of the isolates.

previously identified in Bangladesh; we call these novel mutations in Bangladesh. One mutation in ParC, L88Q, has never been previously identified and documented globally; we refer to it as a worldwide novel mutation. The effects of each mutation have been thoroughly investigated and presented in the following section.

Effects of QRDR mutations in protein-antibiotic interactions

Effects of GyrA mutations

A molecular docking technique has been employed to evaluate the effects of mutations in the GyrA protein. S83L and D87N double mutations affect the optimal binding pose of ciprofloxacin and

Table 1
Analysis of the prevalence of missense mutations in GyrA, ParC, and ParE from the perspective of Bangladesh and outside of Bangladesh.

		GyrA		ParC				ParE	
ID	**Total Mutations**	**S83L**	**D87N**	**S80I**	**E84V**	**E84G**	**L88Q**	**S458A**	**I529L**
U3	2	☑		☑					
U5	5	☑	☑	☑	☑				☑
U6	5	☑	☑	☑		☑	☑		
U8	4	☑	☑	☑				☑	
U10	4	☑	☑	☑				☑	
U15	4	☑	☑	☑				☑	
U17	4	☑	☑	☑				☑	
U21	5	☑	☑	☑	☑				☑
U25	4	☑	☑	☑				☑	
Frequency		100 %	88.9 %	100 %	22.2 %	11.1 %	11.1 %	55.6 %	22.2 %
Identification (Worldwide)		1999	1998	1999	2006	2006	*NA*	2010	2013
Reference		[31]	[32]	[31]	[33]	[33]	*NA*	[34]	[35]
Identification (Bangladesh)		2012	2016	*NA*					
Reference		[36]	[37]	*NA*					

NA: Not Available, not previously identified and documented

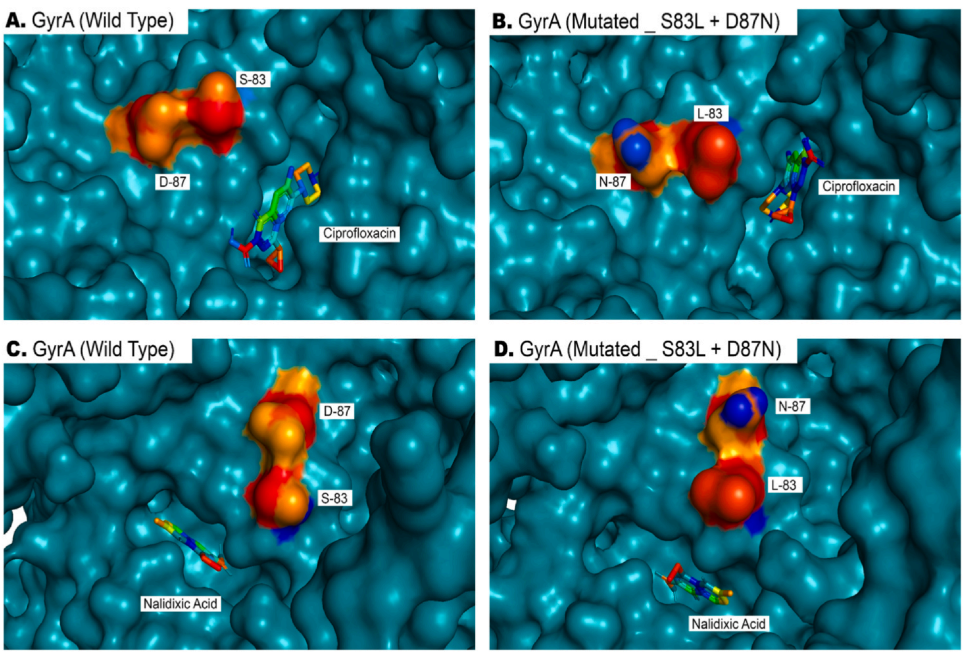

Fig. 3. Docking of ciprofloxacin and nalidixic acid to wild-type and mutated GyrA. *A.* Binding of wild-type GyrA to ciprofloxacin. Default binding conformation of ciprofloxacin to wild-type GyrA. *B.* Binding of double mutated GyrA to ciprofloxacin. The binding conformation of ciprofloxacin has been altered due to the combined effect of S83L and D87N mutations. *C.* Binding of Wild-type GyrA to nalidixic acid. Default binding conformation of nalidixic acid to wild-type GyrA. *D.* Binding of double mutated GyrA to nalidixic acid. The binding conformation of nalidixic acid has been altered and compromised due to the combined effect of S83L and D87N mutations.

nalidixic acid to GyrA (Fig. 3A-3D). The binding conformation of levofloxacin, moxifloxacin, and ofloxacin was unaffected by S83L and D87N double mutations. At the same time, the PQBA of mutated GyrA for ciprofloxacin and levofloxacin was significantly reduced compared to wild-type GyrA. Supplementary Table 5 lists some of the changes in PQBA caused by S83L and D87N double mutations in GyrA.

Effects of ParC mutations

We have identified several combinations of ParC mutations that affected the interaction between the ParC protein and the fluoroquinolone class of antibiotics, particularly with levofloxacin and moxifloxacin (Fig. 4), and significantly reduced the predicted PQBA (Supplementary Table 6). E84V and L88Q double mutations ensue in the altered binding site of levofloxacin within the structure of ParC (Fig. 4B). The site and conformation of moxifloxacin binding are affected by the L88Q single mutation in ParC (Fig. 4D). Other combinations of mutations like S80I and E84G double mutations and S80I, E84V, and L88Q triple mutations affect the optimal binding conformation of moxifloxacin to ParC (Fig. 4E, F). No combination of these mutations in ParC affects the binding conformation or PQBA of ParC for nalidixic acid and ofloxacin.

Effects of ParE mutations

Two mutations in ParE, namely S458A and I529L, were identified, which have never been documented in Bangladesh and have not been analyzed for their effects on ciprofloxacin binding to ParE. A single mutation in ParE, I529L, drastically reduces the binding affinity of ciprofloxacin to the mutated ParE (Supplementary Table 7) and also prevents ciprofloxacin from binding to its optimal conformation with ParE (Fig. 5). S458A alone does neither affect the binding of fluoroquinolones nor reduce the binding affinity (not shown). However, a combination of S458A and I529L mutations exhibits a similar effect as the I529L single mutation, indicating that the S458A mutation probably does not affect fluoroquinolone resistance.

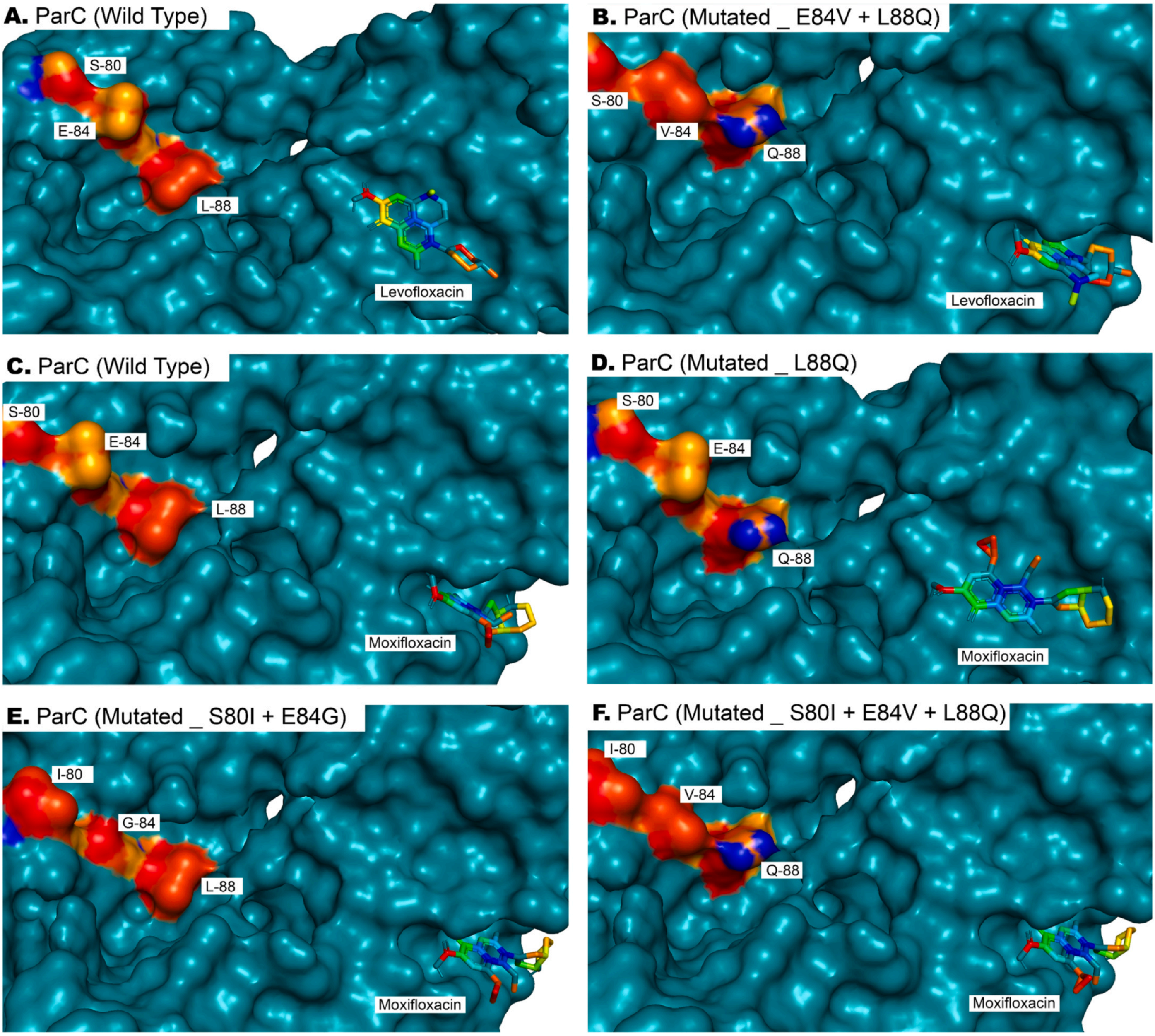

Fig. 4. Docking of levofloxacin and moxifloxacin to wild-type and mutated ParC. *A.* Binding of wild-type ParC to levofloxacin. Default binding conformation of levofloxacin to wild-type ParC. *B.* Binding of double mutated ParC to levofloxacin. The combined effect of E84V and L88Q mutations prevents levofloxacin from binding to its default target site. *C.* Binding of wild-type ParC to moxifloxacin. Default binding conformation of moxifloxacin to wild-type GyrA. *D.* Binding of L88Q mutated ParC to moxifloxacin. The effect of L88Q single mutation prevents moxifloxacin from binding to its default site of target binding. *E.* Binding of double mutated ParC to moxifloxacin. The combined impact of S80I and E84G mutations prevents moxifloxacin from binding to its default site in optimal conformation. *F.* Binding of triple mutated ParC to moxifloxacin. The effect of S80I, E84V, and L88Q also alters the optimum binding conformation of moxifloxacin.

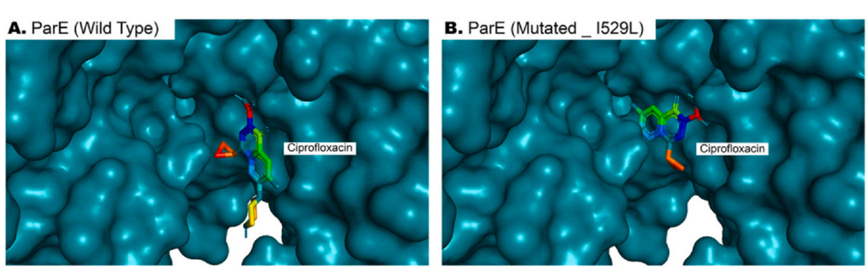

Fig. 5. Docking of ciprofloxacin to wild-type and mutated ParE. *A.* Binding of wild-type ParE to ciprofloxacin. Default binding conformation of ciprofloxacin to wild-type GyrA. *B.* Binding of GyrA with I529L single mutation to ciprofloxacin. The binding conformation of ciprofloxacin has been altered due to the effect of the I529L single mutation.

Table 2
MIC determination of 25 isolates against five fluoroquinolones and their relative severity.

	CIP	LEV	MXF	OXF	NA
Sample	MIC (p-value) / R/S	MIC (p-value) / R/S	MIC (p-value) / R/S	MIC (p-value) / R/S	MIC (p-value) / R/S
U1	0.25 (0.9585) / S	0.25 (0.9444) / S	0.25 (0.9577) / S	1 (0.9528) / S	32 (0.8233) / R
U2	32 (0.4035) / R	4 (0.899) / R	2 (0.9404) / R	16 (0.8166) / R	32 (0.8233) / R
U3*	8 (0.9037) / R	2 (0.9262) / R	4 (0.9129) / R	32 (0.4857) / R	32 (0.8233) / R
U4	32 (0.4035) / R	16 (0.5401) / R	16 (0.4955) / R	16 (0.8166) / R	32 (0.8233) / R
U5*	32 (0.4035) / R	32 (0.0730) / R	32 (0.0388) / R	64 (0.0381) / R	128 (0.0387) / R
U6*	128 (0.0001) / R	64 (0.0001) / R	64 (0.0001) / R	128 (0.0001) / R	256 (0.0001) / R
U7	16 (0.7925) / R	4 (0.8990) / R	8 (0.8241) / R	8 (0.9082) / R	64 (0.4943) / R
U8*	16 (0.7925) / R	32 (0.0730) / R	32 (0.0388) / R	32 (0.4857) / R	64 (0.4943) / R
U9	4 (0.9370) / R	32 (0.0730) / R	32 (0.0388) / R	2 (0.9479) / S	32 (0.8233) / R
U10*	32 (0.4035) / R	16 (0.5401) / R	8 (0.8241) / R	32 (0.4857) / R	64 (0.4943) / R
U11	2 (0.9495) / R	0.25 (0.9444) / S	16 (0.4955) / R	0.25 (0.9561) / S	8 (0.9400) / S
U12	0.25 (0.9585) / S	0.5 (0.9421) / S	0.25 (0.9577) / S	0.25 (0.9561) / S	16 (0.9123) / S
U13	1 (0.9548) / R	0.5 (0.9421) / S	4 (0.9129) / R	0.25 (0.9561) / S	64 (0.4943) / R
U14	2 (0.9495) / R	0.25 (0.9444) / S	16 (0.4955) / S	0.25 (0.9561) / S	32 (0.8233) / R
U15*	32 (0.4035) / R	16 (0.5401) / R	16 (0.4955) / R	16 (0.8166) / R	32 (0.8233) / R
U16	0.25 (0.9585) / S	0.25 (0.9444) / S	1 (0.9509) / R	1 (0.9528) / S	0.25 (0.9589) / S
U17*	32 (0.4035) / R	32 (0.0730) / R	32 (0.0388) / R	32 (0.4857) / R	64 (0.4943) / R
U18	0.25 (0.9585) / S	0.25 (0.9444) / S	1 (0.9509) / R	2 (0.9479) / S	64 (0.4943) / R
U19	1 (0.9548) / R	0.25 (0.9444) / S	2 (0.9404) / R	1 (0.9528) / S	4 (0.9506) / S
U20	0.25 (0.9585) / S	0.25 (0.9444) / S	32 (0.0388) / R	2 (0.9479) / S	4 (0.9506) / S
U21*	64 (0.0174) / R	64 (0.0001) / R	64 (0.0001) / R	64 (0.0381) / R	128 (0.0387) / R
U22	0.25 (0.9585) / S	0.5 (0.9421) / S	0.25 (0.9577) / S	0.25 (0.9561) / S	32 (0.8233) / R
U23	8 (0.9037) / R	0.25 (0.9444) / S	0.25 (0.9577) / S	1 (0.9528) / S	64 (0.4943) / R
U24	16 (0.7925) / R	0.25 (0.9444) / S	0.25 (0.9577) / S	1 (0.9528) / S	64 (0.4943) / R
U25*	32 (0.4035) / R	32 (0.0730) / R	16 (0.4955) / R	8 (0.9082) / R	32 (0.8233) / R

* Isolates sequenced for QRDR mutations; CIP, Ciprofloxacin; LEV, Levofloxacin; MXF, Moxifloxacin; OXF, Ofloxacin; NA, Nalidixic acid; R, Resistant; S, Sensitive; MIC, Minimum inhibitory concentration (μg/mL); p-value: significance of the likelihood of each MIC value being higher than other.

QRDR mutations and fluoroquinolone resistance

We evaluated the impact of novel and rare mutations on antibiotic resistance by comparing the MIC of isolates with specific mutations with those lacking these mutations. For instance, isolates U5 and U21, which possess the ParC E84V and ParE I512L double mutations, exhibited significantly higher ($p < 0.05$) MIC values for ciprofloxacin, levofloxacin, moxifloxacin, and ofloxacin compared to isolates without these mutations. The ParC L88Q+E84G double mutation also resulted in a significantly higher ($p < 0.05$) MIC for all five quinolones tested. However, despite their rarity in bacterial isolates, the ParC S80I and ParE S458A mutations did not significantly increase antibiotic tolerance. These findings (Table 2) support and strengthen the reduced antibiotic-protein binding affinity interpretation.

Discussion

MDR *Escherichia coli,* previously reported to be associated with clinical conditions such as diarrhea [24] and identified in significant quantities in hospital wastewater in Bangladesh [38], is now the predominant bacterial strain responsible for causing UTIs, a prevalent bacterial infection [4]. Antibiotic-resistant strains of these bacteria have emerged as a result of the overuse and improper use of antibiotics in the treatment of UTIs and other infectious diseases [39]. When antibiotics are used to treat UTIs, certain bacteria may survive and acquire antimicrobial resistance by acquiring resistance genes or DNA mutations. Then, these resistant bacteria can then spread to other people through direct contact or the environment, such as tainted water or food. Additionally, the use of antibiotics in livestock husbandry and agriculture has aided in the emergence and spread of bacteria resistant to antibiotics, particularly those that might result in UTIs [40]. Our study comprised only *E. coli* strains; as depicted in Fig. 2B, all of those are multiple drug-resistant. The high prevalence of isolates resistant to five or six different classes of antibiotics, including macrolides, fourth-generation fluoroquinolones, and third-generation β-lactams, is a sign of serious health risks for the residents of Dhaka, Bangladesh.

Results from our experiment reveal that several antibiotics, such as erythromycin, ampicillin, azithromycin, vancomycin, and cefixime, had little to no effect on the UTI isolates. The fact that, despite being a broad-spectrum third-generation cephalosporin, cefixime, which was previously proven to be very effective in treating UTI, respiratory diseases, and some sexually transmitted diseases (STD), was utterly ineffective against 80 % of UTI isolates, is very alarming (Fig. 1). These highly resistant strains can pass on their ability to evade antimicrobials to other microbial communities through horizontal gene transfer, rendering these antibiotics ineffective in treating UTIs [41–44], just like what we observed from our study for ampicillin and erythromycin, against which our sample set was almost resistant. Moreover, these strains can lead to the development of multiple drug-resistance scenarios [41,45–47].

Vancomycin is traditionally ineffective against Gram-negative bacteria like *Escherichia coli* due to their intrinsic resistance, attributed to the outer membrane acting as a permeability barrier. Consequently, no CLSI or EUCAST breakpoints currently exist for interpreting vancomycin susceptibility in *E. coli*. However, emerging evidence suggests that *E. coli* may exhibit increased sensitivity to vancomycin under specific conditions, such as reduced environmental temperatures [48]. In this study, we observed that 20 % of the tested *E. coli* isolates demonstrated susceptibility to vancomycin despite the intrinsic resistance typically observed. This finding prompted us to consider potential temperature-related effects on vancomycin activity and isolate specific factors influencing permeability or drug uptake. These results, while preliminary, underscore the need for further exploration into the interplay between environmental factors, bacterial physiology, and antibiotic efficacy. Understanding the mechanisms driving vancomycin sensitivity in *E. coli* under low-temperature conditions could open avenues for repurposing vancomycin as a subinhibitory treatment, particularly in UTIs where localized environmental factors may enhance drug effectiveness. These findings suggest an intriguing basis for further experimental and clinical studies to clarify vancomycin's activity against Gram-negative pathogens in non-standard conditions.

We utilized a phenotypic approach for carbapenemase detection by using the modified Hodge test, a widely recognized method for assessing functional carbapenemase activity. This assay detects the enzymatic hydrolysis of carbapenems, serving as an indicator of carbapenem resistance in bacterial isolates. Although the Hodge test does not provide definitive identification of carbapenemase-producing isolates, it remains a reliable phenotypic screening tool for detecting carbapenemase activity. For beta-lactamase gene detection, a genotypic approach using PCR was chosen, as the presence of such genes is a definitive marker of beta-lactam resistance with a higher degree of precision. Combining these complementary methods ensured the most optimal and reliable detection of resistance mechanisms, balancing phenotypic functionality and genotypic certainty. The resistance against macrolides and β-lactam antibiotics can be attributed to the widespread use of these antibiotics for many years, which has led to selective pressure on bacteria to develop resistance mechanisms [49,50]. This is precisely why, except for meropenem, other β-lactams and macrolides have severely compromised sensitivity (Fig. 2). Regarding disc diffusion antibiogram and subsequent carbapenemase activity assessment, our investigation reports severe increases in meropenem resistance over the past several years, as evidenced by the significant rise in resistance observed in our study (36 %) compared to the 13 % resistance rate of *E. coli* reported in a 2016 study [51]. Multidrug resistance results from this resistance trait combined with resistance to additional antimicrobial classes. According to the classical definition [1], all of the isolates exhibited the resistance property of MDR. In most cases of our study, any isolate resistant to only three classes of antimicrobials were resistant to macrolides and β-lactams.

Colistin exhibited substantial sensitivity to UTI isolates during the whole susceptibility analysis. According to reports, however, the use of colistin can lead to clinical conditions such as kidney issues, neurological problems, anaphylaxis, muscle weakness, etc., as well as chromosomal mutations that contribute to the emergence of antimicrobial resistance [52,53]. One of these isolates demonstrated extreme resistance to colistin (MIC > 512 µg/mL), which is alarming due to its rarity, as such high levels of resistance are uncommon, and the MIC significantly exceeds previously reported cases of colistin resistance [54]. The alarming colistin resistance, as well as methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin-resistant Enterococci (VRE), ESBL-producing and carbapenem-resistant Enterobacteriaceae has emerged as a significant public health threat, especially in pediatric healthcare settings [55].

All *E. coli* isolates should theoretically possess *gyrA, gyrB, parC,* and *parE* as essential housekeeping genes. However, variations in experimental detectability have been reported, possibly due to mutations in primer binding sites, partial degradation of DNA, or other technical limitations, and have previously been reported to appear in less frequent abundance [24,56]. Furthermore, mutations in QRDRs can alter the primer binding sites, affecting the efficiency of PCR amplification and leading to potential detection failures [57,58]. In our study, non-detection of these genes in specific isolates is likely attributed to experimental and genetic factors rather than their absence. Interestingly, any isolate resistant to more than two of the fluoroquinolones was resistant to all five, even though more than half of the isolates were resistant to all five (Fig. 2D). Resistance to fluoroquinolones is attributed to crucial mutations acquired in the QRDRs of *gyrA*, *gyrB*, *parC*, and *parE* genes [59]. Although these genes are part of the bacterial genome, they aren't always expressed; they are expressed only in actively dividing cells. The high prevalence of these genes indicates that the isolates were actively dividing and targeted by quinolones. These genes may not express at all or express themselves at low levels without selective factors. These genes' expression levels may be increased to provide the resistance phenotype in certain circumstances. Mutations in these genes, however, can confer antibiotic resistance when bacteria are exposed to drugs that inhibit DNA replication and repair. Mutated versions of these genes have slightly different 3D structures, which leads to compromised drug binding and efficacy.

Quinolone resistance and QRDR mutations vary across infection types, with higher prevalence in UTI isolates due to prolonged quinolone use, driving the selection of resistant strains [60]. Diarrhea-causing pathogens, such as *E. coli* O157:H7, may acquire resistance genes less frequently due to differences in colonization niches or host-microbiota interactions. Studies highlight that QRDR mutations, particularly in *gyrA* and *parC*, are more common in UTI isolates than in other infections [61,62]. Plasmid-mediated quinolone resistance (PMQR) genes (i.e., *qnrA*, *qnrB*, *qnrS*) frequently co-occur with QRDR mutations, enhancing resistance levels and facilitating horizontal gene transfer. In Bangladesh, 95 % of UTI pathogens showed phenotypic resistance to lomefloxacin, primarily due to QRDR mutations in *gyrA* [3]. Similarly, high frequencies of QRDR mutations and PMQR determinants are reported in *K. pneumoniae* and *E. coli* nosocomial infections [63]. Biofilm production also increases resistance by shielding bacteria, with a notable association between biofilm formation, *qnr* genes, and *gyrA* mutations in UPEC isolates [64]. Genetic mechanisms, environmental factors, and antibiotic usage patterns contribute to resistance, emphasizing the need for surveillance and stewardship programs to combat its spread.

Our study emphasizes QRDR-mediated resistance but acknowledges the significance of PMQR mechanisms, including *qnr* genes, aac(6′)-Ib-cr, and efflux pumps, in amplifying antibiotic resistance. PMQR facilitates low-level resistance, often as a precursor to high-level resistance when combined with QRDR mutations [65]. Future studies would incorporate a broader analysis of PMQR and efflux mechanisms to understand resistance dynamics fully. Efflux pumps from RND, MATE, and MFS superfamilies significantly contribute to resistance by actively expelling antibiotics, as seen in *Acinetobacter baumannii* and *E. coli* [66]. PMQR genes, found in many *E. coli* and *K. pneumoniae* isolates, further enhance resistance, particularly in conjunction with QRDR mutations like *gyrA* and *parC* [67]. Such dual mechanisms and nosocomial infections are prevalent in the community, underscoring the need for integrated research and surveillance. Additionally, non-genetic factors, including small molecule modulators, warrant further exploration to address resistance comprehensively.

Mutations in the QRDRs of the *gyrA* gene have long been associated with decreased sensitivity of fluoroquinolones [68,69], which aligns with our docking analysis of ciprofloxacin, levofloxacin, moxifloxacin, ofloxacin, and nalidixic acid with wild-type and mutated GyrA. The binding conformation is altered, and at the same time, the new conformation has a weaker PQBA for the drug, which indicates that the optimal interaction is heavily affected (Supplementary Table 5).

ParC mutations had a similar effect to GyrA mutations, decreasing the effectiveness of fluoroquinolones, such as levofloxacin, moxifloxacin, and ciprofloxacin. Our attention was drawn to one particular mutation, L88Q in ParC, and more analysis revealed that this single mutation significantly impacts the overall sensitivity and PQBA of antibiotics. However, the information in Supplementary Table 7 and Fig. 4 also shows another intriguing finding: the catastrophe brought on by the mutation in ParC residue 88 is somewhat lessened by mutations in residues 80 and 84.

Additionally, Supplementary Table 7 lists some impacts of the I529L and S458A mutations in ParE. Notably, I529L alone is detrimental enough to decrease the PQBA of ciprofloxacin and moxifloxacin for ParE protein, and the conjugation of this mutation with S458A has no worse effect than I529L alone. Specific mutations in ParC and ParE and their differential impact on quinolone resistance bear helpful information on the molecular mechanisms involved in antibiotic resistance. ParC E84V and ParE I512L of the mutations analyzed showed clear associations with higher MIC values for several fluoroquinolones, which indicates that these changes significantly impact drug-target interactions. One of the most significant observations is that the ParC L88Q+E84G double mutant exhibited a synergistic effect by conferring increased resistance to all tested quinolones. Such a finding confirms previous reports on the additive effect caused by multiple changes in the QRDRs concerning the resulting resistance patterns.

Interestingly, the novel ParC S80I and ParE S458A mutations did not significantly alter antibiotic susceptibility. This observation underlines that not every mutation in resistance-associated genes may contribute to the resistance phenotype, reflecting the complexity of resistance mechanisms. These findings have important clinical implications for molecular diagnostics and antimicrobial susceptibility testing, suggesting that targeted screening for specific mutations, particularly the ParC E84V and ParE I512L combinations, might better predict quinolone resistance than general mutation screening.

The identification of the novel mutation L88Q in *parC* and other rare mutations in QRDRs (*parC* S80I, E84V, E84G; *parE* S458A, I529L) contributes to the global understanding of fluoroquinolone resistance and underscores the value of molecular surveillance in regions with limited resources. These studies are essential for guiding empirical therapy and developing targeted interventions for UTIs. The parC E84V and parE I512L mutations were also associated with higher fluoroquinolone MIC values. In contrast, the *parC* L88Q+E84G double mutant showed a synergistic effect, conferring enhanced resistance to all tested quinolones, which warrants further clinical investigation.

Our study, conducted at DMCH, provides a comprehensive overview of antimicrobial resistance in Bangladesh, as DMCH serves as a major referral center for patients nationwide. To contextualize our results, we compared them with data from other hospitals in Bangladesh. For example, a recent conducted at a tertiary care hospital in Dhaka found that 71.9 % of uropathogens were MDR *E. coli* isolates [70]. Similarly, a study from Cumilla Medical College Hospital reported significant antimicrobial resistance among uropathogens [71], while another study in Dhaka observed that 37.73 % of *E. coli* isolates were MDR, with high resistance to azithromycin and aztreonam [72]. A report from a tertiary medical college hospital also noted 14 % carbapenem resistance among *E. coli* isolates, despite high sensitivity to amikacin in 90 % of cases [73]. These findings reflect a widespread and concerning antimicrobial resistance among *E. coli* isolates causing UTIs across Bangladesh. They highlight the urgent need for enhanced nationwide surveillance and strategic interventions to address the growing public health threat of antimicrobial resistance.

The findings of this study underscore the complex relationship between genetic alterations and drug-microbe interactions, particularly in the context of *E. coli* resistance patterns in UTIs. The increasing prevalence of multidrug resistance among UTI isolates highlights the urgent clinical and public health implications. The emergence of such resistant strains is primarily driven by the indiscriminate use of antibiotics in treating UTIs [74] and other infections, as well as novel mutations in QRDR [75], which are suspected to drive resistance further. These mutations contribute to fluoroquinolone resistance, potentially enhancing bacterial survival [74] and promoting the spread of resistant strains. Moreover, these mutations may lead to cross-resistance with other antibiotics, further complicating treatment options and escalating healthcare costs.

To address this escalating resistance, the need for antibiotic stewardship is critical. Measures such as restricting unnecessary fluoroquinolone prescriptions, optimizing dosing regimens, and using narrow-spectrum antibiotics where possible are essential steps in curbing resistance. Surveillance programs should also be strengthened to monitor resistance trends and inform treatment protocols. In resource-constrained settings, personalized antibiotic therapies, guided by molecular diagnostics, could improve treatment outcomes. Public health interventions aimed at educating the public on proper antibiotic usage and strengthening resistance monitoring are vital for mitigating the spread of resistant strains. These findings emphasize the need for tailored treatment protocols and the exploration of innovative therapeutic approaches to manage the evolving resistance landscape effectively.

Conclusion

In this study, bacterial isolates from patients with UTIs exhibited high resistance levels to multiple antibiotic classes, particularly β-lactams and fluoroquinolones. Several isolates harbored mutations in QRDRs of *gyrA*, *parC*, and *parE*, with novel and rare mutations identified in ParC (L88Q, S80I, E84V, and E84G) and ParE (S458A and I529L), which had not been previously reported in the context of Bangladesh. Notably, the L88Q mutation in ParC was predicted to alter PQBA and binding conformation, potentially contributing to increased fluoroquinolone resistance. Additionally, double mutations (S83L+D87N) in GyrA and specific combinations of mutations in ParC and ParE were associated with a predicted reduction in fluoroquinolone binding efficiency and elevated MICs. However, while our computational analysis suggests a possible role of these mutations in resistance development, additional phenotypic investigations of the mutants by deleting these identified genes are warranted for validating their roles in the development of quinolone resistance. This study contributes to understanding the molecular landscape of fluoroquinolone resistance. It underscores the importance of nationwide antibiotic stewardship programs, enhanced surveillance, and further research into alternative therapeutic strategies to combat the rise of drug-resistant infections.

Abbreviations

AMP	:	Ampicillin
API	:	Analytical Profile Indexing
AZM	:	Azithromycin
C	:	Chloramphenicol
CCA	:	Chromogenic Coliform Agar
CFM	:	Cefixime
CIP	:	Ciprofloxacin
CLSI	:	Clinical and Laboratory Standards Institute
CT	:	Colistin
DMCH	:	Dhaka Medical College Hospital
E	:	Erythromycin
EMB	:	Eosin Methylene Blue
ESBL	:	Extended-Spectrum Beta-Lactamase
EUCAST	:	European Committee on Antimicrobial Susceptibility Testing
icddr,b	:	International Centre for Diarrhoeal Disease Research in Bangladesh
LEV	:	Levofloxacin
MDR	:	Multidrug-resistant
MEM	:	Meropenem
MHA	:	Mueller Hinton Agar
MHB	:	Mueller Hinton Broth
MIC	:	Minimum Inhibitory Concentration
MRSA	:	Methicillin-resistant *Staphylococcus aureus*
MXF	:	Moxifloxacin
NA	:	Nalidixic Acid
NCBI	:	National Center for Biotechnology Information
OXF	:	Ofloxacin
PCR	:	Polymerase Chain Reaction
PDB	:	Protein Data Bank
PDR	:	Pandrug-resistant
PMQR	:	Plasmid-mediated quinolone resistance
PQBA	:	Protein-Quinolone Binding Affinity
QRDR	:	Quinolone Resistance-Determining Region
R	:	Resistant
S	:	Sensitive
STD	:	Sexually Transmitted Disease

SXT	:	Sulfamethoxazole-Trimethoprim
TE	:	Tetracycline
UPEC	:	Uropathogenic *Escherichia coli*
UTI	:	Urinary Tract Infections
VA	:	Vancomycin
VRE	:	Vancomycin-resistant Enterococci
WBC	:	White Blood Cell
WHO	:	World Health Organization
XDR	:	Extensively Drug-Resistant
ZOI	:	Zone of Inhibition

Ethics approval and consent to participate

The Ethical Review Committee of the Department of Biochemistry and Molecular Biology, University of Dhaka, approved the study (Ref. No. BMBDU-ERC/EC/22/005). Informed written consent was obtained from all the study subjects before the samples were collected.

Author contribution

Manik MRK: Study design; Methodology; Investigation; Wet-lab analysis; Data analysis, validation, and result interpretation; Writing of original draft; Review & editing. **Mishu ID:** Conceptualization; Study design; Methodology; Data analysis, validation, and result interpretation; Writing of original draft; Review & editing; Project administration and supervision; Funding acquisition; Lab equipments and resources. **Mahmud Z:** Conceptualization; Study design; Methodology; Data analysis, validation, and result interpretation; Writing of original draft; Review & editing; Project supervision; Lab equipments and resources. **Muskan MN:** Methodology; Investigation; Data analysis and result interpretation. **Emon SZ:** Methodology; Data analysis and result interpretation; Review & Editing.

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Data availability

All data generated or analyzed during this study are included in this manuscript and its supplementary material files.

Declaration of Competing Interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Not Applicable

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jiph.2025.102766.

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